JOHNS HOPKINS
CAREY BUSINESS SCHOOL

Nonlinear Econometrics for Finance
2 credits

BU.232.630.F1

Fridays 8:30 - 11:30 am
January 24th - March 14th

Spring I 2025
HE Campus

Instructor
Angelo Mele

Contact Information
Email: angelo.mele@jhu.edu
E-mail is the easiest way to contact me.

Office Hours
By appointment

Teaching Assistants: see slides for Lecture 1.

Required Texts & Learning Materials
All materials will be posted online on OneDrive with a link on Canvas in the welcome announcement.

- The slides will be posted one or two days before class.
- Academic articles will also be posted, as needed.

Recommended Texts
A suggested, non-mandatory, technical reading is the book Econometrics by Bruce Hansen. A copy of the book is on OneDrive.

The first ten chapters are largely about linear models (for which I will also provide a set of lecture notes). Chapter 13 and Chapter 5 are about the generalized method of moments and maximum likelihood, techniques which will be central to our discussion.

Another suggested, non-mandatory, reading is the book Asset Pricing by John Cochrane. This is a less technical reading than the previous book but has a substantial finance content. An older version of this book is also posted on OneDrive.

Technology Requirements
We will use Python heavily. Please install the software before the beginning of classes. You should already have installed it from your Computational Finance course in the fall. We will use Jupyter Notebooks throughout the semester.

Course Description
Nonlinear Econometrics introduces econometric tools needed to analyze financial data and build state-of-the-art nonlinear financial models. This is an advanced class requiring strong foundations in multivariate calculus, matrix algebra, probability and statistics. The course covers methods of asymptotic (i.e., large-sample) inference in *extremum* (nonlinear) estimation. Among them, particular emphasis is placed on nonlinear least-squares (NLS), the generalized method of moments (GMM) and maximum likelihood (ML) estimation.

Prerequisite(s)
Computational Finance and Linear Econometrics are prerequisites. We will rely heavily on both courses. Complete familiarity with classical methods of inference in linear models – as introduced in Linear Econometrics - is critical to gain complete understanding of this course's nonlinear methods. Programming in Python – as discussed in Computational Finance – will be used heavily throughout.

Learning Objectives
By the end of this course, students will be able to:

1. Evaluate linear econometric models in terms of their statistical fit
2. Evaluate nonlinear econometric models in terms of their statistical fit
3. Evaluate economic theories using linear and nonlinear methods

Attendance
Participants are expected to attend all scheduled class sessions. Failure to attend class will result in an inability to achieve the objectives of the course. Full attendance - and active participation - are required for you to succeed in this course.

Classroom protocol

- All behaviors and communications in class sessions must be professional, civil and compliant with Carey student policies
- Participants are expected to turn off their phones while in class

Assignments

Assignment	Group or individual	Learning Objectives	Weight
3 homeworks	Group	1 (first HW) 2 and 3 (second and third HW)	10% each
3 in-class quizzes	Individual	1, 2 and 3	5% each
Final exam	Individual	1, 2 and 3	55%
Total			100%

Homework (30%): There will be 3 homework assignments, each worth 10% of the final grade. The assignments have a very important pedagogical role. They are designed to check your understanding of the material covered in class by making you work through an array of theoretical and applied problems. You can work on these in groups (maximum 3 people) but you do not have to do so, if you so choose.

Quizzes (15%): There will be 3 in-class quizzes, each worth 5% of the final grade. The quizzes are in weeks 3, 5, and 7 - at the end of class. They will be short tests, with 2 or 3 questions to be solved in about 15 minutes, designed to check your understanding and knowledge of topics covered in previous weeks.

Final Exam (55%): The (cumulative) final exam will be between 2 and 3 hours long.

Regarding coding
You will, sometime, have issues (everybody does). If you do, you can ask the TA for your class (see slides for Lecture 1) but only after doing the following: (1) consulting available Python resources (virtually every question has been addressed online) and (2) asking one (or more) of your peers. In other words, every time you ask a question about coding you should first begin with your question and then add why (1) and (2) above where not helpful. In the absence of (1) and (2), the TA may not answer your query. This is an advanced course and you should begin training yourself to be creative and independent.

Grading
The grade of A is reserved for those who demonstrate extraordinary performance as determined by the instructor. The grade of A- is awarded only for excellent performance. The grades of B+ and B are awarded for good performance. The grades of B-, C+, C, and C- are awarded for adequate but substandard performance. The grades of D+, D, and D- are not awarded at the graduate level. The grade of F indicates the student's failure to satisfactorily complete the course work. For Core/Foundation courses, the grade point average of the class should not exceed 3.35. For Elective courses, the grade point average should not exceed 3.45.

Policy on Generative AI
Academic integrity is a cornerstone of the Carey Business School. Generative artificial intelligence (AI) tools such as ChatGPT are widely available, and these technologies present a number of exciting opportunities in the classroom. In this course, you may use generative AI tools in any assignment (but never on quizzes or on the final exam). Use of AI must be cited, and all professors have access to an AI indicator on "TurnItIn" which will let them know of the extent to which you likely used AI to complete an assignment. For guidance with referencing AI-generated content, please use the following:
MLA Style Center
The Chicago Manual of Style Online
APA Style

Tentative Course Calendar
Instructors reserve the right to alter course content and/or adjust the pace to accommodate class progress. Students are responsible for keeping up with all adjustments to the course calendar.

Week	Topic	Reading	Goal	HW
1	**Introduction to nonlinear econometrics and technical fundamentals**	Class slides	• Finite sample properties of the sample mean: expected value and variance • Asymptotic properties of the sample mean: (1) consistency and (2) asymptotic normality • Slutsky's theorem • Taylor's expansions • Introduction to nonlinear least-squares (NLS)	**First HW assigned**

Week	Topic	Reading	Goal	HW
2	**Nonlinear least squares (NLS)**	Class slides	• Asymptotic properties of NLS: (1) consistency and (2) asymptotic normality • HAC estimation in NLS	
3	**Generalized Method of Moments (GMM) I** *QUIZ 1 at the end of class*	Class slides	• Asset pricing • The Consumption CAPM (CCAPM) (video) • The GMM criterion • Applying GMM to asset pricing models and beyond • Asymptotic properties of GMM: (1) consistency and (2) asymptotic normality (video)	**First HW due** **Second HW assigned**
4	**Generalized Method of Moments (GMM) II**	Class slides	• The exactly identified case • The over-identified case • Optimal weight matrix • HAC estimation in GMM • Test of over-identifying restrictions	
5	**Maximum likelihood (MLE) I** *Quiz 2 at the end of class*	Class slides	• Examples: IID normal case, AR(1), logit • Asymptotic properties of MLE: (1) consistency and (2) asymptotic normality (video)	**Second HW due** **Third HW assigned**

Week	Topic	Reading	Goal	HW
6	**Maximum likelihood (MLE) II**	Class slides	• Volatility models: ARCH/GARCH • The likelihood of volatility models • The asymptotic properties of the ML estimator in these models	
7	**Maximum likelihood (MLE) III** *Quiz 3 at the end of class*	Class slides	• More on volatility estimation • Robustness/efficiency in econometric estimation • Course recap • Quasi-MLE (QMLE) inference, QMLE vs GMM, QMLE vs ML (video)	**Third HW due**
8	**Final Exam**			

Optional Readings & Resources
Any optional and additional reading will be posted on OneDrive.

Carey Business School Policies and General Information

Please note that failure to become acquainted with Carey policies will not excuse any student from adhering to these policies.

Canvas Site
A Canvas course site is set up for this course. Each student is expected to check the site throughout the semester as Canvas will be the primary venue for outside classroom communications between the instructor and students. Students can access the course site at https://canvas.jhu.edu/.

Technical Support
24/7 technical support for questions regarding Canvas, Zoom, and other technical issues is available. Please refer to Carey's Academic Resources webpage for contact information and other details.

Students with Disabilities - Accommodations and Accessibility
Johns Hopkins University values diversity and inclusion. We are committed to providing welcoming, equitable, and accessible educational experiences for all students. Students with disabilities (including those with psychological conditions, medical conditions, and temporary disabilities) can request accommodations for this course by providing an Accommodation Letter issued by Student Disability Services. Please request accommodations for this course as early as possible to provide time for effective communication and

arrangements. For further information or to start the process of requesting accommodations, please contact Student Disability Services at the Carey Business School.

Academic Ethics Policy
Carey expects graduates to be exemplary global citizens in addition to innovative business leaders. The Carey community believes that honesty, integrity, and community responsibility are qualities inherent in an exemplary citizen. The objective of the Academic Ethics Policy (AEP) is to create an environment of trust and respect among all members of the Carey academic community and hold Carey students accountable to the highest standards of academic integrity and excellence.

It is the responsibility of every Carey student, faculty member, and staff member to familiarize themselves with the AEP and its procedures. Failure to become acquainted with this information will not excuse any student, faculty, or staff member from the responsibility to abide by the AEP. Please contact the Office of Student Affairs if you have any questions. For the full policy, please visit the Academic Ethics Policy webpage.

Student Conduct Code
The fundamental purpose of the Johns Hopkins University's regulation of student conduct is to promote and to protect the health, safety, welfare, property, and rights of all members of the University community as well as to promote the orderly operation of the University and to safeguard its property and facilities. Please contact the Office of Student Affairs if you have any questions regarding this policy. For the full policy, please visit the Student Conduct Code webpage.

Commitment to Respect
Respectful behavior creates an environment within the Carey Business School where all are valued and can be productive. Carey defines respectful behavior as conduct that, at a minimum, demonstrates consistent courtesy for others, including an effort to understand differences. As such, all in the community agree to the Carey Commitment to Respect, which states that we all strive to show that we value each other's human dignity and our differences, and to choose behavior and language that demonstrates mutual respect. Please visit the Commitment to Respect webpage to learn more about the expectations and resources available.

Classroom Policies for All On-Site and Remote-Live Classes
Carey is committed to maintaining the highest standards of excellence in all forms of instruction. To that end, we have developed policies and procedures for all classes offered in on-site and remote-live formats. These policies will govern all courses occurring in these formats, and all students are expected to familiarize themselves with and adhere to these policies.

Student Success Center
The Student Success Center offers assistance in core writing and quantitative courses. For more information, visit the Student Success Center webpage.

Other Important Policies and Services
Students are encouraged to consult the Student Handbook and Academic Catalog and Student Services and Resources for information regarding other policies and services. For your convenience, there is a singular website students can visit to learn about all JHU and Carey policies.

Copyright Statement
Unless explicitly allowed by the instructor, course materials, class discussions, and examinations are created for and expected to be used by class participants only. The recording and rebroadcasting of such material, by any means, is forbidden. Violations are subject to sanctions under the Academic Ethics Policy.